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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech (CST) (CBCS) Degree Examinations

December 2024- January 2025

22CST102 – STRUCTURAL DYNAMICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Differentiate Between static and Dynamic Load with examples.	10	L*2	1	1
	b) State and Explain D'Alembert's Principle of Dynamic Equilibrium.	10	L2	1	1
2.	a) What is Degree of Freedom? And how to express degree of freedom in dynamics.	10	L2	2	2
	b) The elevated water tower tank with a capacity for 5000 gallons of water shown in Fig 2. Has a natural period in lateral vibration of 1.0 sec. Determine the lateral stiffness k of the tower and the weight W of the tank. Neglect the mass of the supporting columns(One gallon of water weights approximately 8.34lb)				

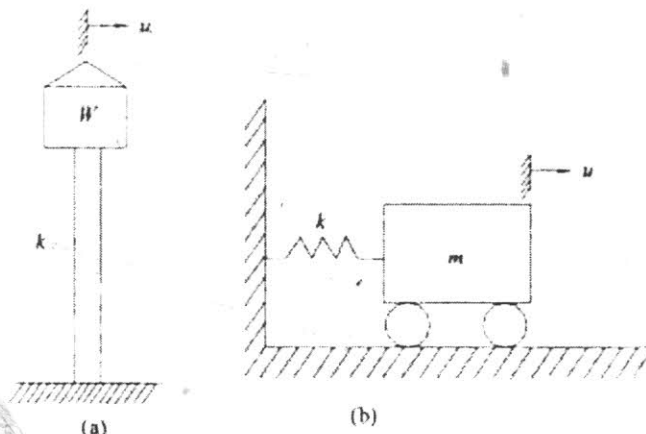


Fig 2

10 L3 2 3

Unit – II

3.	a) Derive the Equation of motion for damped free vibration of SDOF System and also derive equation of motion for critically damped, Over damped and under damped system.	20	L6	2	3
4.	a) A platform of weight $W= 4000$ N is being supporting by equal columns that are clamped to the foundation as well as to the platform. Experimentally it has been determined that a static force of $F=1000$ N applied horizontally to the platform produces a displacement of $\Delta= 0.1$ in. it is estimated that damping in the structure is of the order of 5% of the critical damping. Determine for this structure the following: i. Undamped natural Frequency ii. Absolute damping coefficient iii. Logarithmic decrement iv. The number of cycle and the time required for the amplitude of motion to be reduced from an initial value of 0.1 to 0.01 m.	20	L6	2	3

Unit – III

5. a) Explain formulation of MDOF equation of motion.
b) Determine Natural Frequency and mode shape for the shear building as shown in the figure 5.

05 L3 2 3

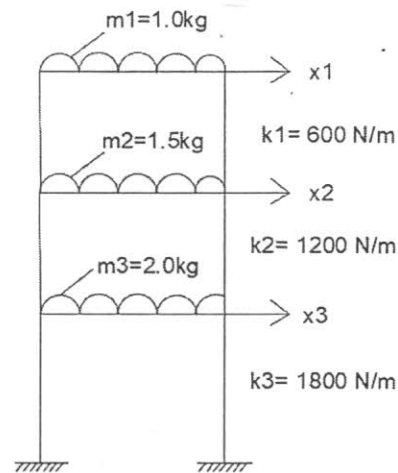


Fig 5

15 L5 3 3

6. a) Explain Free vibration of MDOF System.
b) Explain characteristics of Free vibration of an undamped multiple degrees-of-freedom (MDOF).

10 L3 3 2

10 L3 3 2

Unit – IV

7. a) Explain Forced vibration without damping (SDOF).
b) Explain Rayleigh's method.

10 L5 4 3

10 L3 4 3

8. a) State and Derive Duhamel's integral method.
b) Explain Rayleigh-Ritz method.

15 L5 4 3

05 L2 4 3

Unit – V

9. a) Explain Earthquake Ground motion.
b) Explain the following
A) Seismometer
B) Accelerometer
10. a) Discuss any Two laboratory experiments in dynamics.

10 L4 4 3

10 L5 5 3

20 L4 5 4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
